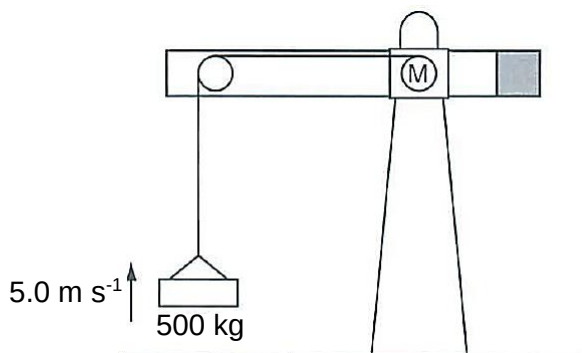


- 6 The motor M in a crane is used to lift a load of 500 kg vertically upwards at a constant speed of 5.0 m s^{-1} . During the lifting of the load, energy is supplied to motor M at a constant rate of $3.0 \times 10^4 \text{ J s}^{-1}$.



Which of the following statement is **not** correct?

- | | |
|----------|--|
| A | The efficiency of motor M is 82%. |
| B | The kinetic energy of the load is 6.3 kJ. |
| C | The work done on the load by gravity is positive during the lifting of the load. |
| D | The rate of gravitational potential energy gained by the load is about $2.5 \times 10^4 \text{ W}$. |

Answer: C

Rate of gravitational potential energy gained,

$$= Fv = mgv = 500 \times 9.81 \times 5.0 = 24525 = 2.5 \times 10^4 \text{ W (D is correct)}$$

$$\text{Efficiency of the motor} = \frac{\text{Power output}}{\text{Power input}} \times 100\% = \frac{24525}{3.0 \times 10^4} \times 100\% = 82\% \text{ (A is correct)}$$

The kinetic energy of the load,

$$\frac{1}{2}mv^2 = \frac{1}{2} \times 500 \times 5.0^2 = 6.3 \text{ kJ (B is correct)}$$

The gravitational force acts downwards on the load while the load's displacement is upwards. As such, the work done by gravity (gravitational force) is negative during the lifting of the load. (C is incorrect)

7	A golf ball on level ground is hit and starts to move at an angle of 60° to the horizontal.
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